



UNIVERSITY OF DHAKA

Real-Time DDoS Detection Using Machine Learning and Flow-Based Network Analysis

Submitted by

Exam Roll:524

Registration No: H-123

Exam Roll:531

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Submitted to the

Department of Computer Science and Engineering

University of Dhaka, Bangladesh

**in partial fulfillment of the requirements for the degree of
Professional Masters in Information and Cyber Security (PMICS)**



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Declaration Form



University of Dhaka

We hereby declare that the work in this report titled "Real-Time DDoS Detection Using Machine Learning and Flow-Based Network Analysis" has been carried out by us under the supervision of Prf. Dr Rezaul Karim We further declare that where information has been derived from the literature, we have adequately cited and referenced the original sources. No part of this report was previously presented for another degree or diploma at this or any other institution.

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"Common sense is like deodorant. The people who need it most never use it."

Bill Murray

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Abstract

istributed Denial-of-Service (DDoS)" attacks are among the most prominent threats to network security, capable of overwhelming systems and causing significant operational and financial damage. This paper introduces a machine learning-based method for flow-level network analysis-based real-time "DDoS" assault detection and categorization. A Random Forest model was trained and evaluated on the S2017 dataset, covering attack types such as DoS hping3 and Slowloris, and further applied to live traffic detection. The model demonstrated consistent performance across multiple train-test splits (e.g., 70-30, 60-40), achieving 100% precision in controlled experiments with zero false positives, although performance in real-world environments may vary, as discussed in the limitations. The suggested method provides a lightweight, scalable, and useful solution for real-time deployment in dynamic network situations by utilizing aspects such as packet counts, byte quantities, and session duration.

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Abstract

"Distributed Denial-of-Service (DDoS)" attacks are among the most persistent threats to network security, capable of overwhelming systems and causing significant operational and financial damage. This paper introduces a machine learning-based method for flow-level network analysis-based real-time "DDoS" assault detection and categorization. A Random Forest model was trained and evaluated on the CICIDS2017 dataset, covering attack types such as DoS hping3 and DoS Slowloris, and further applied to live traffic detection. The model demonstrated consistent performance across multiple train-test splits (80-20, 70-30, 60-40), achieving 100% precision in controlled experiments with zero false positives, although performance in real-world environments may vary, as discussed in the limitations. The suggested system provides a lightweight, scalable, and useful solution for real-time deployment in dynamic network situations by utilizing aspects like packet counts, byte quantities, and session duration.

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Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Background

Distributed Denial of Service (DDoS) attacks have emerged as one of the most persistent and disruptive forms of cybercrime in the modern digital era. These attacks

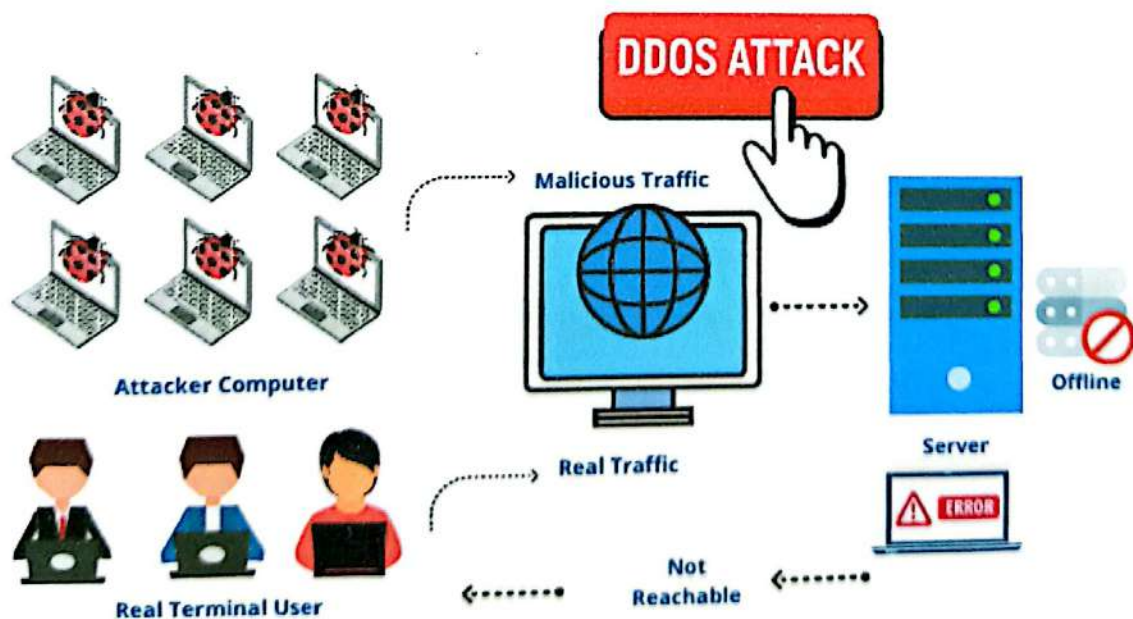


FIGURE 1.1: DDoS Attack

work by overwhelming a targeted system—whether a server, network, or online service—with excessive volumes of malicious traffic. By exhausting essential resources, including bandwidth, memory, and processing power, attackers succeed in avoiding legitimate users from accessing the service. The consequences are severe: prolonged downtime, possible data loss, financial setbacks, and reputational harm for the affected organization. With the rapid expansion of online services and critical digital infrastructures, the frequency and sophistication of DDoS attacks have increased significantly, posing an ongoing and dynamic threat across multiple sectors.

Traditional detection approaches, such as signature-based methods and deep packet inspection (DPI), provide only partial solutions. Signature-based systems depend heavily on predefined attack patterns, making them incapable of recognizing novel or slightly modified attack vectors, such as zero-day threats. DPI, though capable of identifying anomalies in packet contents, demands significant computational resources and struggles in real-time, especially when dealing with encrypted traffic or high-throughput environments. These limitations make it evident that new approaches must be developed to achieve timely and scalable DDoS detection.

Flow-based network analysis, when combined with machine learning (ML), offers a compelling alternative. Instead of inspecting packet payloads, flow-based methods examine metadata from traffic flows—such as packet counts, byte volumes, session durations, and source/destination addresses. These features are both lightweight and protocol-agnostic, making them particularly suitable for real-time monitoring. When integrated with ML algorithms, such statistical representations of network activity provide a robust framework for detecting and classifying malicious traffic patterns, even in dynamic or encrypted environments.

The present research focuses on designing and evaluating a real-time DDoS detection system that leverages flow-based features in conjunction with supervised machine learning models. The CICIDS2017 dataset, a widely recognized benchmark that simulates real-world attack scenarios, was employed for training and testing the models. Three algorithms—Random Forest, XGBoost, and Logistic Regression—were compared based on detection accuracy, false positive rates, and runtime

efficiency. Among these, Random Forest demonstrated superior performance, combining high accuracy with robustness against overfitting, and thus was identified as the most suitable choice for real-time deployment.

We leverage CICFlowMeter (Java 8, cic. cs. unb. ca. ifm. Cmd) for creation of flow-level CSV features compatible with CICIDS2017. To provide live demos, we record with Scapy (Linux) or Wireshark/dumpcap (Windows). Our Windows demo also contains a Python flow table + feature extractor to the form used during training.

In conclusion, this thesis demonstrates that integrating flow-based network analysis with machine learning provides an effective, real-time, and scalable solution for detecting DDoS attacks. By addressing the shortcomings of traditional detection methods, the proposed system contributes to the development of practical and adaptable defenses against the growing spectrum of cyber threats, thereby offering organizations a reliable mechanism to safeguard critical digital infrastructures.[8, 19].

- **Cyber Security and Distributed Denial-of-Service (DDoS) Attacks**

A "Distributed Denial-of-Service (DDoS)" attack is one of the most critical security concerns in modern computer networks. It refers to a malicious attempt to disrupt the normal operation of a system, service, or network by overwhelming it with an excessive volume of traffic. Unlike a conventional Denial-of-Service (DoS) attack, which typically originates from a single machine or a single network source, a "DDoS" attack leverages a large number of compromised devices—commonly mentioned as botnets distributed across multiple geographic locations. These devices simultaneously generate malicious traffic towards the target, making the attack far more difficult to trace and mitigate, as the traffic often appears to come from legitimate sources [11].

The phenomenon of DDoS first drew widespread attention in 1999, when researchers documented an incident at the University of Minnesota. More than 114 compromised networks were exploited using a tool called *Trin00*, which generated millions of packets in a very short time frame. The attack

effectively exhausted the university's network capacity and resulted in a two-day outage [7]. This marked the beginning of large-scale awareness regarding the destructive potential of distributed attacks. Since then, DDoS incidents have grown significantly in scale and sophistication, becoming a serious danger to online services to the reliability and availability of online services, financial systems, and national critical infrastructures [20].

The mechanics of a DDoS attack center around the exhaustion of computational or communication resources at the target system. By directing overwhelming amounts of malicious traffic, attackers can cause server crashes, service unavailability, or even full-scale infrastructure failure. Two widely used techniques in such attacks are reflection and amplification. In reflection-based attacks, requests are spoofed to appear as though they originated from the intended victim. As a result, intermediary servers respond with amplified traffic directed towards the victim, unknowingly participating in the attack. Amplification, on the other hand, exploits network protocols that generate disproportionately large responses compared to the size of the request. Attackers abuse such protocols to magnify their impact, enabling them to cripple a target with relatively little effort [17].

Broadly, "DDoS" attacks can be categorized into three major categories. Volume-based attacks focus on saturating the bandwidth of the target system by generating enormous traffic volumes, often measured in bits per second (bps). A common example involves abusing the UDP protocol, where crafted packets trigger excessive responses from the target, resulting in traffic amplification that overwhelms legitimate users [2]. Protocol-based attacks, by contrast, exploit vulnerabilities in network protocols to exhaust server resources such as CPU cycles, memory, or connection tables. A well-known example is the TCP SYN Flood, which abuses the three-way handshake by sending many SYN requests but never completing the handshake. This leaves the server's connection backlog saturated and unable to handle legitimate requests [10]. Finally, application-layer attacks target services at the application layer (Layer 7 of the OSI model). They are particularly challenging because

they require minimal traffic to cause disruption. The Slowloris attack, for instance, maintains many open connections to a web server by sending partial HTTP requests very slowly. This starves the server of resources while evading traditional traffic-based monitoring [16].

In summary, the evolution of DDoS attacks reflects the growing complexity of cyber threats faced by modern networks. Their ability to undermine services at various levels – e.g. from raw bandwidth usage to protocol abuse and application level targeting, makes these attacks a difficult problem for organizations to solve. By studying the mechanisms of these attacks, and how they affect the network, this thesis intends to provide a valuable input towards an ongoing research effort to design efficient and scalable solutions for detection and mitigation of DDoS.

- **DDoS Attack Cases** Distributed Denial-of-Service (DDoS) attacks have evolved rapidly in the past few years, increasing in magnitude and complexity. These are now asserting themselves as substantial threats to the cyberworld at large, in notable events spanning 1999 to 2025. What began as rather basic floods, have evolved into sophisticated, massive operations that can encompass botnets, amplification, reflexion, etc. Following are some of the important events in the evolution of DDoS attacks:

Trin00 Botnet – University of Minnesota (1999):

The first DDoS attack that the general public is aware of took place in 1999 when the Trin00 botnet attack the University of Minnesota in a coordinated assault. The attack employed more than 114 compromised networks to create millions of packets that resulted in heavy network congestion and two days of downtime. Turning point This incident was an eye-opener, illustrating the destructive capabilities of a distributed attacker and launching botnets as a formidable force in cyberspace. [12].

Yahoo! Attack (2000):

In February 2000, Yahoo! – still the internet's leading web portal – was hit with a massive DDoS attack that caused its services to become once unavailable for hours on end. It was early days in DDoS development, and such a large-scale attack on a commercial enterprise was a clear sign of a new threat to business operations and a company's brand. It paved the way for cyberattacks to become more commercialized and have a real-world impact [6].

The Mirai Botnet (2016): A major turning point in the history of distributed denial-of-service (DDoS) attacks came with the release of the Mirai botnet in 2016. The malware took advantage of poorly secured Internet of Things (IoT) devices—such as home cameras and routers—to assemble a massive botnet. One of its most infamous operations was the attack on the DNS provider Dyn, which reached nearly 1.2 Tbps of traffic. The disruption caused widespread outages across popular platforms including Twitter, Netflix, and Reddit. Mirai demonstrated how IoT exploitation could be combined with large-scale automation, setting a new standard for DDoS campaigns.

GitHub Memcached Attack (2018): In February 2018, GitHub experienced what was then the largest DDoS attack ever recorded, peaking at 1.35 Tbps. The attackers used a technique called memcached amplification: small requests were sent to misconfigured servers, which then produced disproportionately large responses aimed at GitHub. Although the company's defenses absorbed the assault within minutes, the incident highlighted the risks of poorly configured internet services and illustrated just how powerful amplification techniques had become.

Amazon Web Services (2020): Two years later, Amazon Web Services reported that it had successfully mitigated a 2.3 Tbps DDoS attack—the largest publicly acknowledged at the time. The attack relied on CLDAP reflection, a method that generates highly amplified responses from vulnerable servers. This case not only underscored the persistence of reflection-based attacks but also emphasized the growing threat facing cloud providers, whose infrastructure supports a significant portion of the global economy.

Cisco Router Exploitation – Salt Typhoon Campaign (2024–2025): Between late 2024 and early 2025, the threat group known as Salt Typhoon—linked to China—exploited critical flaws in Cisco IOS XE (CVE-2023-20198 and CVE-2023-20273). More than a thousand devices worldwide were compromised, including systems in the U.S., South Africa, Italy, and Thailand. The attackers installed backdoors and tunneling mechanisms to maintain long-term access, raising concerns that the same footholds could be used to launch DDoS operations. Cisco responded with urgent security patches, particularly for vulnerabilities in its IOS XR software that could trigger denial-of-service conditions. This campaign showed how router-level exploitation can enable both espionage and large-scale disruption.

Cloudflare Record Attack (2025): In mid-2025, Cloudflare announced that it had stopped the largest DDoS attack ever observed. A single IP address was hit with 37.4 terabytes of data in only 45 seconds, peaking at 7.3 Tbps. The attackers relied on UDP reflection and amplification, further demonstrating how DDoS attacks have grown not only in size but also in speed. The scale of this event reflects both the sophistication of offensive techniques and the mounting pressure on defenders tasked with keeping critical systems online.

Geopolitical Campaigns and Law Enforcement Response (2023–2025): During this period, pro-Russian hacktivist groups such as NoName057(16) conducted a wave of politically motivated DDoS operations. Their targets included European government agencies, banks, energy firms, and transport networks—an effort to disrupt essential services and apply geopolitical pressure. In response, Europol coordinated Operation Eastwood, which led to arrests, server seizures, and the dismantling of key parts of the group's infrastructure. This case illustrates the increasing role of DDoS in geopolitical conflicts and the critical importance of international cooperation in mitigating such threats.

The transformation of DDoS attacks from basic flooding to complex and broad attacks mirrors the growing power of cyber attackers. Now, these attacks take advantage of insecure IoT equipment, misconfigured server systems and even infrastructure, compromising not only commercial businesses but also governmental

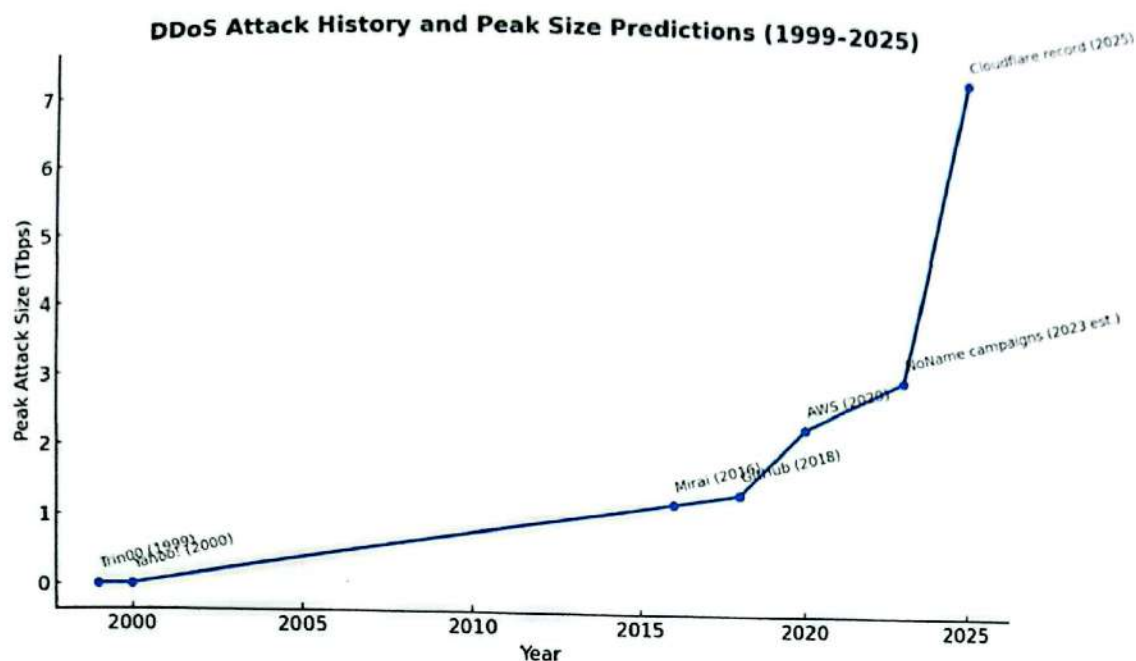


FIGURE 1.2: DDoS Total Attack History and Predictions 2025 graph

facilities. As cyber threats are constantly developing, the demand for strong defense, cooperation and pro-active risk management is increasing.

1.1.2 Motivation

Bangladesh is nowhere lagging behind in the faster wave of digital transformation, more reliance on mobile finance, government portals and online services. But this advancement means an increased risk to cybercrimes, especially DDoS attacks. Such attacks can cause significant interruptions to the flow of finance, communication, and vital services, and thus are a concern for the country's IT infrastructure.

Bangladesh has faced a spike in cybercrime, with banks, mobile money clients, government websites and other utilities coming under DDoS attack in the past few years. In fact, even agencies like the e-Gov CIRT and the National Cyber Security Agency have warned of DDoS attacks targeting essential services. Public confidence and economic security The worrying increased attacks online, notably during nettension moments – like pay day – represent a real threat to both public confidence and the national economic stability.

Although worldwide research for DDoS detection utilizing machine learning (ML) offers promising results, these solutions frequently only be tested offline and in a laboratory setting. In addition, applied use of these models in the field in resource-limited settings, such as Bangladesh, are limited. The conventional DDoS discovery methods are too high cost or less suitable for domestic infrastructure, so they proved to be ineffective for most heterogeneous Bangladeshi network.

Our work addresses these challenges by developing a localized, real-time DDoS detection system that should be readily adoptable in the country by adapting to Bangladeshi traffic patterns. We have demonstrated a real-time pipeline interference to Random Forest and XGBoost ML models and hope to propose a lightweight and adaptive solution to secure the critical digital infrastructure of Bangladesh. It isn't just a DDoS detection project, it is a real-time prevention one, and for banks, government services and other critical services, it's also about keeping the lights on.

By doing so, we minimize the knowledge chasm between education and industrial implementation, transforming the proved DDoS detection models work to a ready made solution for Bangladesh's high risk high traffic environment.

1.2 Related Work and Gaps in Current DDoS Detection Approaches

A number of existing work and systems have examined the application of ML for DDoS detection; however, they mostly target offline and controlled scenarios rather than real-time and deployable solutions. In the work of Sharafaldin et al. (2017) [18] and Alkasassbeh (2020) [1] have released datasets such as CICIDS2017 and CICDDoS2019 for ML models to be trained on detecting DDoS. Although these works achieve high accuracy almost 99% on distinguishing DDoS attacks, they only evaluate the models on static datasets without showing their performance on real traffic traces.

Apart from research efforts, several commercial measures such as costly foreign appliances and content delivery networks (CDNs) have been used to counter DDoS attacks. These solutions are not, however, tailored to local infrastructure and are not feasible for Bangladesh's specific network environment, particularly considering their cost-effectiveness and real-time nature. With their potential to overload computational and communication resources, DDoS attacks have consistently emerged as one of the most severe threats to network security. These attacks are constantly changing and growing in complexity, size and methods of attack. The continuum of DDoS attack holes, from traffic flooding to protocol abuse and application-layer attacks, overburdens conventional defenses which are stuck to an out-of-date protection model [2].

Most of the existing systems, such as signature-based intrusion detection systems (IDS) and static threshold-based approaches fail to effectively detect new or low-rate DDoS attacks. This is due the fact that they are primarily based on pre-defined rules and history based attack patterns, lost the ability to defend previously unknown attack approaches. Moreover, such solutions often incur a high false positive rate, which degrades the performance of network monitoring and incident response. This is especially problematic in scenarios in which prompt detection and response is essential. [2].

A further problem of the prior art DDoS detectors is that they cannot be used in real-time. Today's DDoS attacks can rapidly find their way to terabit-scale traffic levels in a matter of seconds, giving organizations little time to respond. Flow-based monitoring solutions, which only examine packet headers instead of whole packets, offer a lighter solution to monitor the traffic and can scale to large networks. Although it is more efficient than full payload inspection, the traditional flow-based anomaly detection methods usually do not possess the level of adaptive learning to distinguish legitimate traffic surge, flash crowds and malicious traffic floods [2].

In view of these challenges, machine learning (ML) has been considered as a potential candidate to enhance the accuracy and flexibility of DDoS detection systems. Using ML models, networks are able to learn complex traffic patterns depending

on a number of flow features, like packet inter-arrival times, byte counts, and flow durations and generalize them to detect previously unseen attacks. ML, however, it is still a challenging task to introduce ML to real-time DDoS detection mechanisms due to the following reasons: 1) Low computational cost, 2) Scalability (to be applicable to high rates of network traffic), 3) Avoid making false alarms (after all, high accuracy is preferred to unnecessary alerts). [2].

The main research question considered in this work is How can be a real-time DDoS detection system designed and implemented that is based on machine-learning for flow-based network analysis. The goal is a system which has high detection accuracy and low false positive rates and is scalable for operations. This is especially important in protecting contemporary computing networks, such as cloud computing infrastructure and critical facilities such as power grids where the reliable availability of services is essential.[2].

Additionally, while many solutions proposed, like those proposed in CIC-IoT 2023 dataset address the IoT-based botnets, they have not yet managed to develop a deployable and real-time system to operate in the resource constrained high-risk regions including Bangladesh [13].

1.3 Research Objectives

This research mainly focuses on how to combine machine learning and flow-based network analysis to design and implement the real-time detection framework for DDoS attacks. The key objective of this research is to develop a large-scale, high-fidelity and flexible defense mechanism that is able to protect the modern network infrastructures more efficiently than the legacy detections.

The specific aims of the study area:

Characterizing network flows: Explore flow-level aspects of the network (i.e., packet size distribution, inter-arrival times, flow duration and protocol distribution, etc.)

TABLE 1.1: Research Problems in Existing DDoS Detection and Our Proposed Solutions

Research Problem	Our Proposed Solution
Limited Real-World Deployment	Integration of feature extraction into a real-time capture pipeline using Wireshark/dumpcap, enabling live traffic analysis and immediate threat detection.
Static Datasets Used for Training	Adaptation to Bangladesh-specific traffic patterns, including BDIX, university networks, and mobile money peak loads, for accurate detection in local contexts.
No Real-Time Mitigation of DDoS Attacks	Automatic triggering of alerts, logging of incidents, and firewall rule application to actively mitigate attacks in real time.
High Complexity and Inflexible ML Models	Use of a model Random Forests (lightweight and interpretable) to balance performance, accuracy, and practicality.
Ineffective Solutions for Local Infrastructure	Design of a lightweight, locally deployable system that can operate effectively within Bangladesh's network infrastructure constraints.
Expensive, Inefficient Foreign Solutions	Cost-effective ML-based system tailored for local networks, avoiding reliance on expensive foreign appliances or CDNs.
Lack of Targeted Solutions for Critical Sectors	Customized protection for banks, mobile finance platforms, government portals, power, and utility services, ensuring continuity during peak usage or attacks.

to discover representative patterns that differentiate between benign and malicious activities.

Contribute to the development of machine learning models: Developing and training models for detecting multiple threats, including volumetric, protocol-based, and application-layer DDoS attacks, based upon flow-level data.

Develop a real-time detection architecture: Develop an architecture capable of detecting with low latency and minimal computation but maintains high detection accuracy and low false positives in changing network environments.

Performance analysis: A complete analysis is carried out on the proposed framework with benchmark intrusion detection datasets and simulated real-time traffic

to measure detection rate, false positive rate, precision, recall and processing time.

Compare to prior art: You must benchmark a system proposed against traditional solutions, such as signature-based detection, threshold based-detection, and so on, to show that it is able to achieve adaptation, scalability and resistance, in respect to newer and newer attack strategies.

Propose practical deployment: Propose practical deployment methods applicable to the real world, especially focusing on cloud systems, enterprises, and critical services where service availability and operational continuity is key.

Chapter 2

Literature Review

2.1 Machine Learning Approaches for DDoS Detection

Time-based characteristics derived from network traffic flow data have proven to be valuable in identifying and categorizing Distributed Denial of Service (DDoS) attacks. In a study by Halladay et al. (2022) [9], they showed that by using a 25-feature time-based subset, they achieved over 99% accuracy in binary classification and more than 70% accuracy in multiclass classification across 12 types of attacks. This approach offers a significant advantage by reducing both training time and computational complexity while still providing accuracy comparable to using a full-feature dataset. This conclusion highlights the power of time-related features to develop scalable, real-time DDoS detection systems.

Breiman (2001) [3] broke the solid ground in machine learning when he introduced Random Forests. Random Forests is an ensemble method that aggregates the predictions of several decision trees to increase classification and regression accuracy. It achieves higher generalization by combining predictions of multiple trees, particularly suitable for large and high-dimensional datasets. One advantage of Random Forests is the valuable information it provides on feature importance, which is beneficial in problems where we need to interpret the model.

XGBoost (Chen and Guestrin, 2016) [5] is another powerful tool from the machine learning community. It is a scalable gradient boosting framework that makes decision tree learning efficient, which is good for tasks such as DDoS attack detection. XGBoost is an interesting candidate due to its cutting-edge capabilities, including L1 and L2 regularization frameworks, sparsity-aware algorithms, and efficient support for large-scale unbalanced datasets, which are common in network traffic analysis problems. One of its major strengths is that it can deal with missing values, making it suitable for real-time DDoS detection systems and greatly enhancing training speed and accuracy.

There are two major reasons for the importance of the proposed taxonomy: it gives a clear bird's-eye view of the entire landscape through its multiple dimensions and is a cornerstone for designing efficient detection systems. This framework classifies the attacks according to their type, attack tools, and effects on network security, paying particular attention to the migration of volumetric attacks to application-layer attacks. This transition emphasizes the necessity of adaptive and real-time detection approaches, like the machine learning methods proposed in this work, to enhance detection performance and mitigate different types of attacks.

A DDoS cloud botnet study reveals that cloud systems face specific vulnerabilities, as compared to traditional networks, and are more susceptible to DDoS attacks because of their distributed nature and multi-tenant model [14]. It categorizes DDoS attacks into three types: volumetric, protocol, and application layer. This research highlights the urgency of developing real-time detection tools for cloud-based networks, especially machine learning-based ones, to efficiently defend against DDoS attacks, which fits well with the goal of this research—to make detection and classification of DDoS attacks in such dynamic and distributed environments more efficient.

The study on machine learning and deep learning-based IDS has proven their abilities to address large-scale network traffic and be adaptive to unknown attack patterns. These approaches are suitable for real-time detection systems as they can enhance size efficiency, intercept more DDoS information, and balance detection

accuracy. Furthermore, deep learning has demonstrated its merits in dealing with high-dimensional data and detecting latent patterns, which are critical to enhancing the scalability and accuracy of IDSs.

Finally, we examine machine learning for network intrusion detection methods, considering the issues of deploying such approaches in practice. The research highlights the limitations of classical rule-based systems in the presence of DDoS, a dynamic and advanced attack. It also emphasizes the importance of considering feature selection and model interpretability for developing high-performing and trustworthy ML-based IDS. Such insights are clearly relevant to our project's goal of using machine learning to detect and discriminate DDoS attacks in an accurate and scalable fashion [15].

Chapter 3

Dataset

3.1 Data Analysis

3.1.1 The CICIDS2017:

Dataset has become a widely used benchmark in intrusion detection research due to its realistic design and comprehensive coverage. It was created to simulate the day-to-day traffic of a corporate network, combining typical user activities—such as web browsing, email, file transfers, and VoIP calls—with a variety of staged attack scenarios. This approach produces a dataset that more closely reflects real network behavior than older synthetic benchmarks, while still providing accurate ground-truth labels suitable for supervised learning. The included attacks cover a broad spectrum, from high-volume floods such as DoS Slowloris and distributed DoS attacks, to brute-force attempts on SSH and FTP services, web-based exploits like SQL injection and cross-site scripting, and subtler activities such as infiltration and botnet communication. The dataset also contains rare examples, including the Heartbleed SSL exploit, ensuring diversity across both common and uncommon threats. **Class Imbalance:** A notable characteristic of CICIDS2017 is the pronounced imbalance between classes. Some attack types, particularly flooding-based attacks, dominate the dataset with millions of flows, whereas others—like Heartbleed or

infiltration—appear only a few dozen times. Benign traffic is heavily represented, as expected. This imbalance presents a challenge: a naïve classifier could achieve seemingly high accuracy by favoring majority classes while failing to detect rare but critical attacks. In essence, CICIDS2017 reflects a real-world truth: most network traffic is benign, and the most dangerous attacks are often rare and stealthy. Consequently, careful evaluation metrics—such as per-class recall and F1-scores—are essential for meaningful model assessment.

- **Features After Preprocessing:** After cleaning and pruning, approximately 60 numerical features were retained. These features capture multiple perspectives of network behavior:
 - Time-based characteristics: flow duration, idle periods, and active intervals.
 - Traffic volume indicators: bytes and packets per flow in both forward and reverse directions.
 - Transport-layer behaviors: flag counts, average segment size, and flow bytes per second.
 - Application-layer statistics (where available), such as HTTP request/response sizes. The raw dataset contains many redundant or highly correlated features. By removing constants and pruning correlations above 0.95, the feature space was simplified without losing representational diversity. The resulting feature set is compact yet expressive, capable of describing both high-volume attacks and subtle, low-profile probes.

3.1.2 Observed Challenges:

Despite its advantages, CICIDS2017 presents several challenges:

- Class imbalance, which can obscure rare attacks.
- Temporal clustering, as attacks were launched in bursts on specific days, necessitating careful train-test splitting to avoid leakage.

- Presence of synthetic artifacts; for example, scripted DoS floods behave differently from adaptive real-world adversaries.
- Flow aggregation, which can mask packet-level details important for detecting stealthy exfiltration or protocol misuse.

Nonetheless, the dataset strikes a practical balance: it is sufficiently large to train modern machine learning models, diverse enough to evaluate generalization, and widely used, enabling comparability across studies. To improve consistency, Lycos-based preprocessing was applied to address labeling issues reported in prior work, producing a manageable and reliable benchmark for evaluating intrusion detection systems.

3.2 Dataset Description

The dataset employed in this work is the CICIDS2017 dataset, developed by the Canadian Institute for Cybersecurity (CIC). It has become a widely adopted benchmark for intrusion detection research due to its unique combination of realism and diversity. Unlike earlier synthetic datasets, CICIDS2017 was collected in a controlled but realistic enterprise testbed, consisting of servers, routers, switches, and client machines that generated both benign and malicious traffic.

The benign portion of the dataset reflects common day-to-day activities within an organizational network. It includes web browsing over HTTP/HTTPS, email exchanges using SMTP, POP3, and IMAP, file transfers through FTP and SFTP, remote sessions via SSH, voice-over-IP (VoIP) calls, and routine DNS queries. This diversity ensures that the “normal” traffic profile is rich enough to represent the baseline behavior of enterprise networks.

3.2.1 Benign Traffic Types:

The benign class includes legitimate network activities that simulate everyday usage across different protocols and services. These traffic types are essential for establishing a baseline of normal network behavior.

- **Normal Browsing:** Standard web-based activities that emulate human interaction with web servers.
- **HTTPS Traffic:** Secure web access generated through TLS/SSL protocols, reflecting encrypted communication.
- **SSH Sessions:** Remote terminal access and administrative operations over secure shell connections.
- **FTP Transfers:** File download and upload sessions, representing traditional client-server data exchange.
- **DNS Requests:** Domain name resolution traffic, a fundamental component of nearly all Internet-based communication.

Collectively, these benign traffic categories capture common network services and enable the model to learn baseline characteristics across diverse ports and protocols.

3.2.2 Attack Traffic Types

The attack traffic in the dataset comprises well-known Distributed Denial-of-Service (DDoS) and web-based attacks. These scenarios provide diversity in both volumetric and stealthy techniques, allowing comprehensive training of the detection model.

Slowloris: A low-bandwidth, stealth-based DoS attack that maintains numerous open connections by sending partial HTTP requests, thereby consuming server connection capacity. **hping3:** Flooding attacks scripted with hping3 and other tools exhibit patterns that are more uniform than real-world adversarial traffic.

These attack types collectively represent volumetric, protocol-based, and application-layer threats, making them suitable for building a robust and generalizable detection model.

Rationale for Dataset Selection

We used a merged CICIDS2017 CSV across days; we also regenerated/refined labels with Lycos and aligned features before training due to the following attributes: **Realism:** The traffic reflects typical workday network activity, enhancing the validity and applicability of results. **Label Quality:** Flows are accurately labeled as BENIGN, DoS Hulk, and DoS Slowloris, which is critical for supervised machine learning. **Attack Diversity:** The dataset includes both high-volume floods and low-rate stealthy attacks, enabling generalization across diverse strategies. **Balanced Composition:** The relatively even distribution of benign and malicious flows reduces class imbalance during training.

Chapter 4

Methodology

4.1 Work Flow and Model Analysis

Random Forest (RF) is a supervised machine learning model that performs implicit feature selection through embedded techniques by determining the number of features considered at each tree node. RF, applicable to both supervised and unsupervised tasks, relies on ensemble learning, combining multiple decision trees to enhance predictive performance and reduce overfitting. For classification tasks, RF generates multiple decision trees during training and aggregates their outputs to improve accuracy. The model's prediction can be formulated as:

$$RF(x) = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

where $RF(x)$ is the predicted output for input x , N is the number of trees, and T is the prediction of the i -th tree. Figure 3 illustrates the Random Forest architecture with N decision trees.

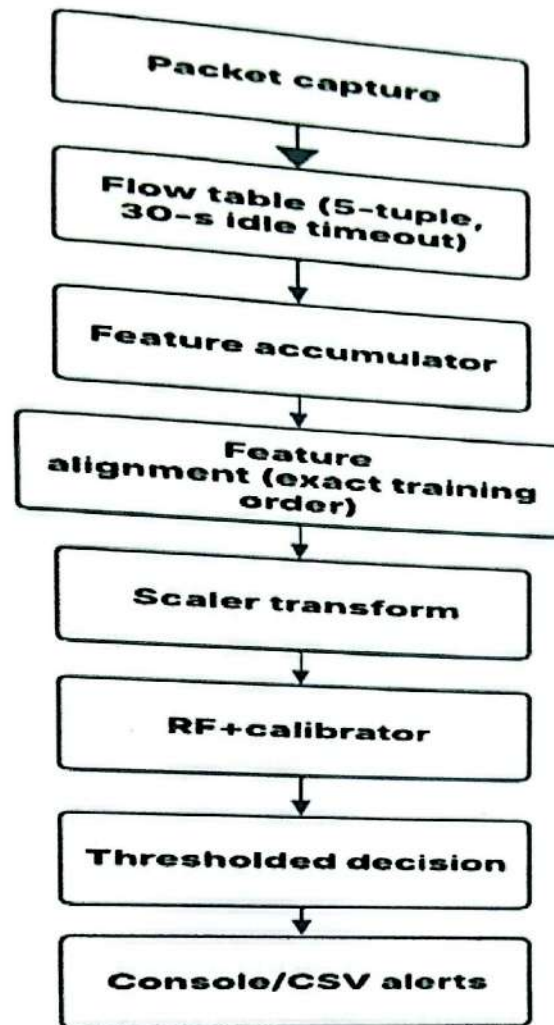


FIGURE 4.1: Workflow with Dataset

4.2 Model Description

The Random Forest (RF) model is a robust ensemble-based machine learning algorithm that has been extensively applied in intrusion detection research. It belongs to the family of bagging (bootstrap aggregating) methods and operates by constructing a collection of decision trees during training. Each tree votes independently, and the majority decision is selected as the final prediction. This approach significantly reduces the risk of overfitting compared to individual decision trees, making Random Forest highly suitable for complex network intrusion detection tasks. In this work, the Random Forest classifier is applied to the Wednesday-workingHours dataset, utilizing 60 extracted flow features such as packet lengths, flow durations, source and destination ports, protocol-specific statistics, and TCP

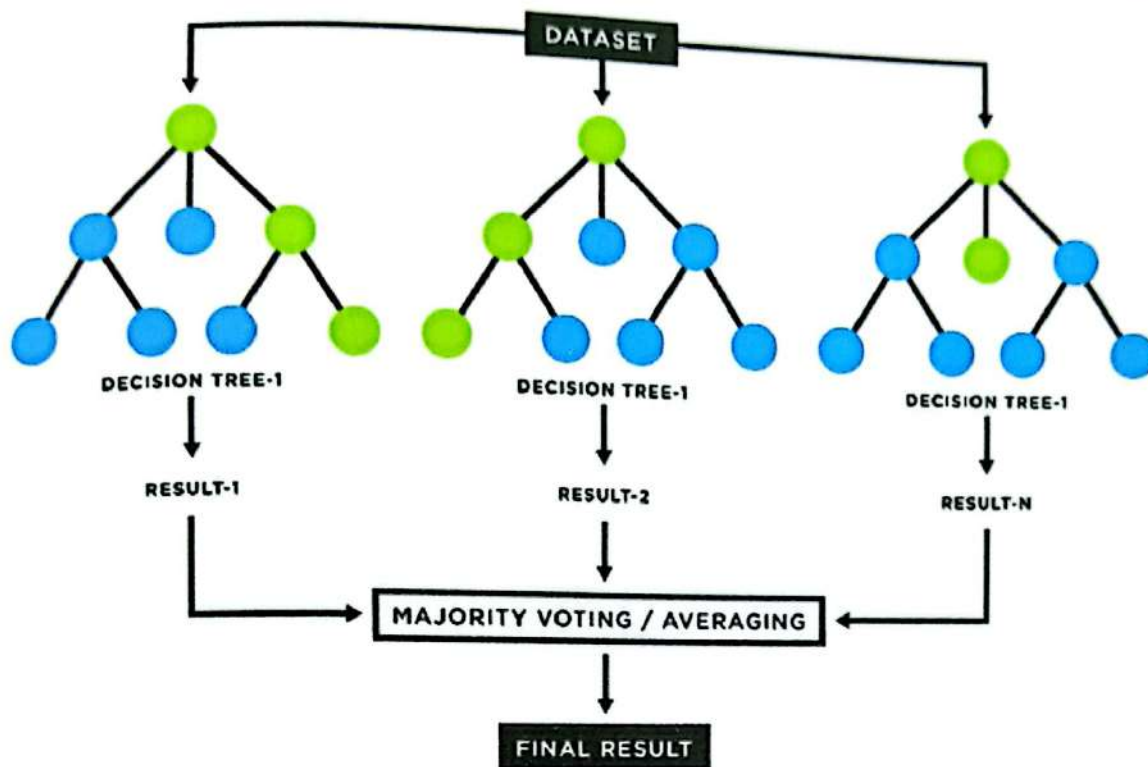


FIGURE 4.2: Random Forest Architecture

flag counts. These features capture diverse aspects of traffic behavior, which help distinguish between benign traffic and malicious DDoS flows. By leveraging multiple decision trees trained on different bootstrap samples of the dataset, Random Forest can effectively capture non-linear relationships and diverse attack signatures within the data. One of the strengths of Random Forest lies in its ability to handle imbalanced datasets. By aggregating predictions from multiple trees, the classifier improves detection of underrepresented attack classes, ensuring better sensitivity to minority DDoS variants. Furthermore, Random Forest provides feature importance rankings, which are valuable in identifying the most influential traffic attributes for attack detection, aiding explainability in security research.

To evaluate model stability and generalization, three different train-test splits were adopted: 80–20, 70–30, and 60–40. During training, the ensemble of decision trees learned traffic flow patterns indicative of DDoS attacks, including anomalies in packet size distributions, abnormal flow durations, and irregular port usage.

The performance of the model was measured using standard classification metrics—accuracy, precision, recall, and F1-score—to ensure balanced evaluation.

Comprehensive testing demonstrated that Random Forest achieved consistently high detection performance across different splits. The best-performing configuration was selected for potential real-time deployment. The final model was capable of predicting not only whether a DDoS attack occurred but also distinguishing among different attack variants with strong reliability. Its scalability, interpretability and robustness against overfitting render Random Forest a suitable and powerful solution for real-time dynamic network intrusion detection.

4.3 Model Training and Evaluation

1) Data Splitting SPLITS Metric (a) 60-40 (b) 70-30 (c) 80-20 As for the train and test splits of the data the overall dataset is horizontally split into 3 configuration sets to determine the random separations and also to assume that the model generalizes well for the new network packets makes use of a different subsets. By considering the model performance across these alternative partitions, one can gain a sense of the model's sensitivity to the data distribution and a more reliable indication of its likely performance in real-world use. Such a methodology enables us to know the behavior of the model under different data conditions, so the results are more reliable.

2) Base Learner and Training Strategy: Our baseline classifier is the Random Forest model (approximate configuration: 800 trees; at least 2 samples per leaf; log2 feature selection on each split). (a) Uneven distribution: The CICIDS2017 dataset is highly skewed where benign network traffic is dominant, and some attack class instances are underrepresented. To handle this issue, we used a class-weighted technique with data-rebalancing:

Undersampling of benign flows: This was done to prevent the benign class from dominating the classifier's learning process.

Oversampling of minority attack classes: Specifically, web-based attack types were oversampled to ensure that these less frequent, but significant, attack categories were adequately represented.

These strategies ensure that the classifier learns to treat all classes equitably, avoiding the common pitfall of neglecting rare but critical attack types, which is crucial in cybersecurity applications where identifying all attack vectors is vital.

3) Random forest calibration and threshold optimization models, while powerful, often produce predicted probabilities that are poorly calibrated, which can negatively impact decision making in real-world applications. To enhance the reliability of the predictions, we applied isotonic calibration to the predicted probabilities. This step was crucial for improving prediction accuracy and was performed using either a pre-fitted model or Stratified K-Fold Cross-Validation, depending on the number of samples per class.

Instead of using a single threshold across all classes, we performed threshold optimization on a per-class basis during cross-validation. This optimization aimed to maximize the macro-averaged F1-score, which balances the precision and recall across all classes, with particular attention to rare classes where a uniform threshold could miss critical attacks. The final threshold configurations were saved in a `class_thresholds.json` file, ensuring that the process is fully reproducible for future experiments.

4) Evaluation Metrics and Reporting To assess the model's performance more systematically, we tested the model on both cross-validation and a held-out test set, using a variety of standard classification metrics:

Precision (P), Recall (R), and F1-Score: Their values were estimated not only class-by-class but also averaged upon the classes for showing the performance of the model for different traffic categories.

Confusion Matrices: These were useful visualizations to finding patterns in misclassification, and seeing where the model was having trouble or bias.

Histograms of Max Predicted Probabilities: This visualization showed the model's certainty around predictions, such a model being confident in its predictions and confidence in being unsure of them.

Furthermore, a comprehensive report was created for each experiment, reporting on essential parameters, calibration results, and threshold settings. This text and PDF report helps ensure the evaluation process is both accessible and reproducible for the research community.

4.4 Real-Time Detection and Scalability

To assess the model's feasibility for real-time deployment, the following measures are implemented: 1. **Simulated Traffic:** Real-time traffic simulations are performed to evaluate the model's responsiveness to varying traffic patterns. This simulation includes both benign and malicious DDoS traffic, helping assess the model's efficiency in distinguishing between legitimate and attack traffic in dynamic network conditions. 2. **Scalability Testing:** The system's scalability is rigorously tested by gradually increasing the traffic volume. The model's performance under higher throughput scenarios is evaluated to ensure it can handle large-scale network traffic without significant degradation in detection accuracy or processing time. 3. **False Positive Rate:** Minimizing false positives is a crucial component of real-time detection. The model's ability to avoid flagging legitimate traffic as an attack is critical in reducing unnecessary mitigation actions and minimizing disruptions in network services.

4.5 Deployment Considerations

The world we live in, with all the so much important happens that has a place only in an ideal world, wherein each packets that has been exchanged is some how treated to be trusted. Key deployment considerations include:

- **Cloud Infrastructures:** Built for the cloud, system was designed with scalability and reliability in mind. It is able to accommodate the varying traffic pattern in dynamic cloud environments, providing seamless service.
- **Enterprise Networks:** In essential fields such as healthcare and financial services, where consistent service is needed, the model allows for the detection and response to attacks in real time. This reduces the downtime in which critical services are knocked offline.
- **Critical Services:** In essential fields such as healthcare and financial services, where consistent service is needed, the model allows for the detection and response to attacks in real time. This reduces the downtime in which critical services are knocked offline.

4.6 Evaluation and Comparison

The Random Forest model's performance is assessed and compared with Support Vector Machine (SVM), XGBoost, and Logistic Regression in the following. The comparison is made with respect to:

- **Detection Accuracy:** This criteria evaluates how accurately the model can distinguish between benign and attack traffic. The model is reliable in that the attacker are easily detected and false negative false positive are low on the classification of normal traffic. Comparisons among models will give more details concerning which model will work more efficiently in terms of detection accuracy for diversified DDoS attacks.
- **Scalability:** The scalability test helps verify that the model can scale with the increased amount of data in high-throughput network condition without deteriorating the performance. The system has to stay fast as the load grows. By comparing the models, we can see that one of them has better capability for large-scale real-time traffic.

• **Robustness:** This evaluates the capability of the model to cope with different DDoS attack types, such as volumetric, protocol-based, and application-layer attacks. The well train model is resilient to various attack vectors and it is able to generalize to the new attack and unknown threat model. The comparison will also evidence which model can best catch up-and-coming new DDoS attacks. This evaluates the capability of the model to cope with different DDoS attack types, such as volumetric, protocol-based, and application-layer attacks. The well train model is resilient to various attack vectors and it is able to generalize to the new attack and unknown threat model. The comparison will also evidence which model can best catch up-and-coming new DDoS attacks.

• **Real-Time Applicability:** Monitoring speed and real-time detection are really necessary to mitigate the influence from DDoS. In this paper, we will assess the models' ability to detect attacks quasi-instantaneously without causing a heavy computational overhead or slowing the decision making time up to infeasible limits. Such a comparison will show the most responsive model with high detection accuracy.

By conducting performance comparison of Random Forest with SVM, XGBoost, and Logistic Regression, the merits and drawbacks of each model in detection accuracy, scalability, robust and real-time will be presented. The results will pave the way for practical deployment in the network. cting DDoS

4.7 Model Choice

Randomforest: This output is from a Random Forest model predicting on a holdout test set. Both classes (0 and 1) have perfect precision, recall, and f1-scores of 1.00. The classification report shows that for both classes, the model predicts with 100 percent accuracy, as indicated by the "accuracy" score of 1.00. The macro average and weighted average also have scores of 1.00, confirming the overall perfect

performance of the model. The support values (88007 and 88006) indicate the number of instances in each class within the test set.

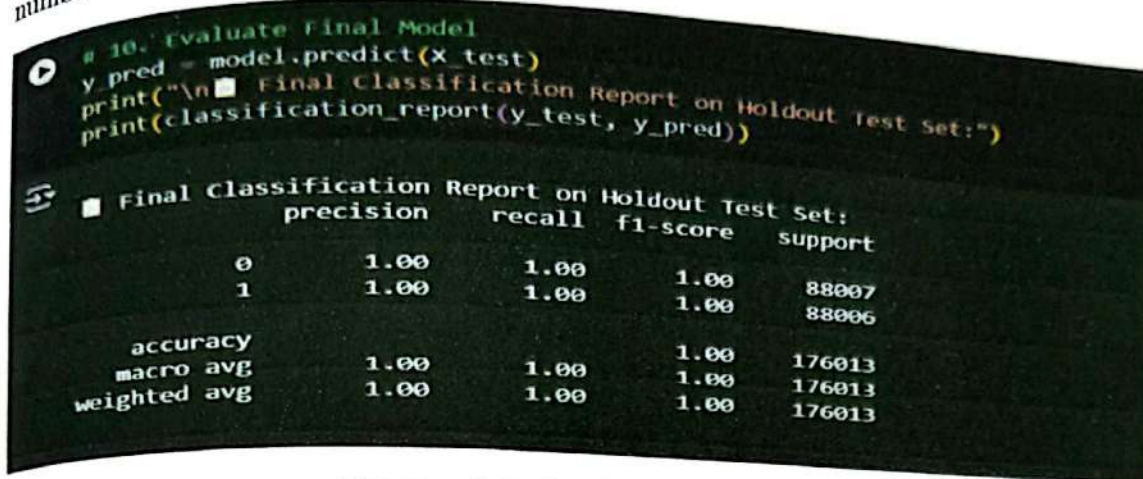


FIGURE 4.3: Random Forest Result

XGBoost: This report comes from an XGBoost model, showing similar results to the first. It again reports perfect precision, recall, and f1-scores of 1.00 for both classes. The accuracy, macro average, and weighted average scores are also all 1.00, indicating flawless performance on the test data. The support values are identical to the first report, suggesting that the test set contains the same number of instances for each class.

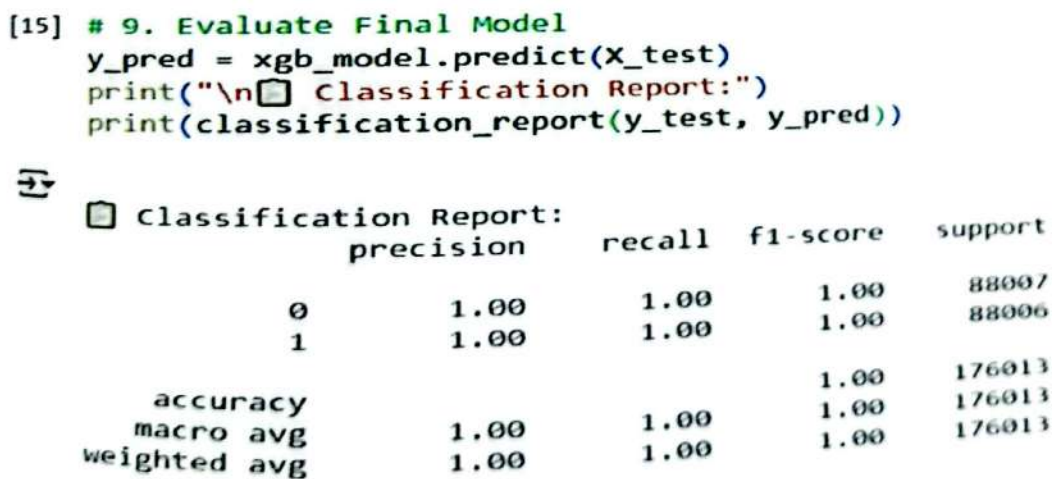


FIGURE 4.4: XGBoost Result

Logistic Regression: In this case, the model's performance has slightly decreased compared to the first two. The precision, recall, and f1-scores are now 0.95 for both classes (0 and 1), still indicating strong performance but with a slight reduction. The accuracy, macro average, and weighted average are all 0.95, implying that

the model's overall performance is slightly less than perfect but still very good. The support values remain the same, reflecting the unchanged size of the test set for each class.

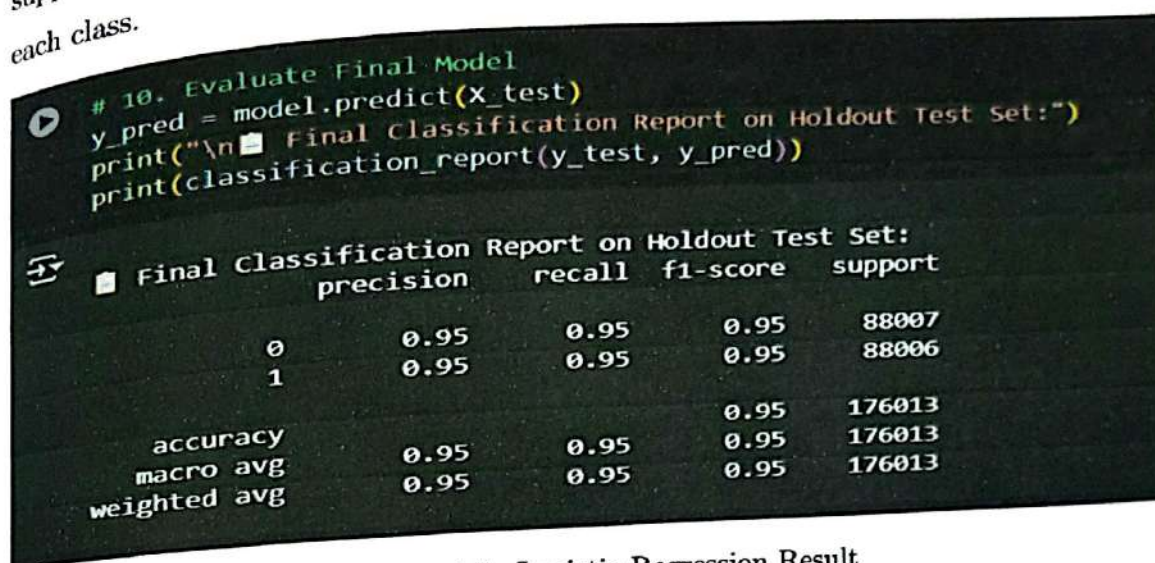


FIGURE 4.5: Logistic Regression Result

The performance of the three supervised classifiers—Random Forest, XGBoost, and Logistic Regression—was evaluated on the test dataset using standard classification metrics: Accuracy, Precision, Recall, and F1 Score. The results are summarized in the table below:

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000
Logistic Regression	0.9987	0.9985	0.9988	0.9986

TABLE 4.1: Model Performance Metrics

4.8 Justification for Selecting Random Forest

- **High Accuracy and Robustness:**

The classifications from the Random Forest model is excellent and continues to maintain accuracy on new data. This demonstrates the strong generalization ability of our model, which makes it a good candidate for tasks that need reliable predictions. Furthermore, Random Forest is known for its robustness

and it is not prone to overfitting. It is particularly appropriate for modeling both linear and non-linear relationship in complex data set. And its performance with high-dimensional, complex data makes it an attractive choice for predictive tasks in a wide range of fields [4].

- **Faster Inference Time:**

Random Forest has also faster inference time compared to other models like XGBoost. Especially in real-time applications, the low-latency quality is especially important as the latency for critical systems can bring about significant performance decline. In applications such as cybersecurity threat detection, the need for rapid decision making makes Random Forest's faster inference time beneficial, enabling faster detection, response and leading to lesser lag in response to the security threat [4].

- **Simplicity and Interpretability:**

One of the main advantages of Random Forest is that it is easy to implement. Unlike more complicated models (i.e. XGBoost) that typically require some tuning of parameters, Random Forest is super quick to toss into place, making it an elegant choice for those looking for a "quick and easy" solution. It also offers explainable insights on feature importance, which can help the analysts to better understand the rationale behind the decision of the predictions. This interpretability is essential in scenarios where transparency is required, namely in security applications where one needs to understand how detection decisions are reached [4].

- **Integrability and Low Computational Load:**

Random Forest is a light weight model compared to something like XGBoost. It has less external dependencies, and less preset, so it is lightweight. This property is potentially useful in the context of deploying machine learning models in production, particularly in resource-constrained settings. Its lower computational cost makes it easier to deploy and operate, it cuts down on the infrastructure investment and makes for easier deployment [4].

4.8.1.1 System Overview

The new IDS design is implemented in Python to achieve high flexibility and portability, integrating open-source tools. The system follows a modular architecture and includes several key components:

- **Scapy:** Our tool to capture packets in real time and parse raw network traffic at the data-link layer. This allows the system to efficiently watch the network.
- **BFlow Table:** A dynamic module that groups the captured packets into bidirectional flows. The group is created by using the commonly used 5-tuple features, such as the Source, Destination IP, Source, Destination Port, and the Protocol Type are the important features used to form the group.
- **Feature Extraction Engine:** Facial flow-level statistics does indeed distill key flow-level patterns, such that higherorder patterns can be extracted CICFlowMeter/CICIDS2017 feature schema (counts, sizes, timing, flags) at lower dimensions. This step is essential to convert raw network data into features that can be provided to the machine learning model.
- **Inference Engine:** Loads the pre-trained Random Forest classifier and scaler (again, serialized by joblib), classifies each flow as benign or malicious. The inference engine supports real time classification and decision support.

This modular system allows monitoring described network very efficiently and in real-time behavior anomaly is detected. The architecture is expressed to be easily extended and scaled and the resulting IDS can be applied for dynamic network.

4.8.1.2 Flow Construction

For flow-based analysis, the incoming packets are aggregated to network flows by the common 5-tuple identifier:

FLOW CONSTRUCTION To perform flow-based analysis, incoming packets are classified as flows based on the well-known 5-tuple (srcIP, dstIP, srcPort, dstPort, protocol). Flows are monitored for each flow in a dedicated flow table with an idle timeout of 30s. If no packet is received by a particular flow in this window, the flow is 'teared down' and flushed from the memory.

Each flow is maintained in a dedicated flow table that includes essential metadata, such as timestamps, number of packets, and TCP flags.

Have implemented a timeout, and the idle flows are closed in 30 seconds. If no packets on a flow are seen within a given timeout period, the flow is terminated and the memory is released. This guarantees effective memory utilization clearing out inactive flows.

The flowlevel abstraction results in the less memory overhead compared to storing per-packet behaviour in addition to a better way to execute meaningful statistical measurements. Through the flow-derived attributes, the model could capture the macroscale network dynamics which enable the systems to achieve the optimal performance and scalability for intrusion detection.

4.9 Real-Time Feature Extraction

For each active flow, real-time statistical features are extracted, computed and updated on the fly as new packets arrive. These features are selected to closely approximate the feature set that the model was trained with and which discriminate flow behavior for the given classifier. The main characteristics are then extracted as:

- **Packet Statistics:** Number of all packets in the flow, Forwarding and Reverting packets count, according to the direction of the flow.
- **Byte Statistics:**

Min, max, mean packet size. These statistics are informative of the magnitude and variability of the traffic inside the flow.

- **Temporal Features:**

The average and standard deviation of the difference in arrival time between consecutive packets. (Vincent and Allwein) These features allow us to capture the timing and periodical nature of the traffic, which can be important in anomaly detection.

Flow duration, by determining the time delta between the first and last packet of the flow.

- **TCP Flag Counters:**

Number of packets with PSH and ACK flags set. These features are frequently indicative of scanning or flooding activities observed during network attacks.

Those are efficiently maintained and updated in real-time with Python dictionaries and NumPy arrays. This technique allows feature computation with as low latency and as little resource consumption as possible. The architecture is meant to handle high amounts of network traffic and provide fast and accurate feature extraction, crucial for real-time detection of anomalies. This streamlined feature extraction mechanism is essential for timely responses to potential security threats.

4.9.0.1 Feature Alignment and Model Inference

The features are critically realigned and preprocessed by this step, and only then, they can be input for the classifier at the extracted intermediate flow data:

- **Feature Vector Construction:** The feature vector is then structured in the original order and format as used for training. These step's purpose is to ensure that the features are normalised correctly and the results are the same during generation as in training, and this is crucial to the accuracy of the models.

- **Feature Normalization:** The feature vector is normalized through the trained StandardScaler to have the same scale of the used features during training. This process scales the feature to make them fall within the same range, which is important to for the model to perform well. It can be expressed as the following scaling operation:

$$\text{scaled} = \text{scaler.transform}([\text{extracted_features}]) \quad (4.1)$$

Model Inference: We then classify the scaled feature vector using the pre-trained Random Forest model. This process of inference determines normal or anomalous flow based on the learned patterns. The inference is expressed as:

$$\text{prediction} = \text{model.predict}(\text{scaled}) \quad (4.2)$$

After prediction, Predicted Label and its flow key properties (source and destination IP, source, and destination ports) important for forensic analysis and network monitoring purposes are logged. This streaming model inference process will allow proactive identification of anomalies in network operations for early threat detection and subsequent mitigations.

4.9.0.2 Output Generation and Logging

To support the operation transparency as well as the post-analysis, the system produces two classes of output:

- **Real-Time Terminal Output:** Detected flows are immediately displayed in real-time also on the network terminal. This result instantly reveals the categorization and behaviors of network traffic, such as:
- **Detection Time:** The time of day at which the flow was detected.
- **Classification Label:** Whether the flow is categorized as benign or attack.
- **Source and Destination IP Adresse:** The IP address pair of the flow.

- **Source and Destination Port Numbers :** The port number through which the end points are communicating with.

- **Persistent Logging:**

All detected malicious flows, according to the system, are recorded into a persistent file detections. csv. This log is a verbose log designed to allow you to do a complete and comprehensive forensic analysis, to either prove that you have been hacked, or to find out how you got hacked. Information logged at each log entry Right at each entry in the log is a fixed-sized entry that should not take much more than three or four bytes.

- **Detection Timestamp:** The Time when the Malicious Flow was Detected.
- **Source and destination IP addresses and ports** Source and destination Communication endpoints and its port of the flow.
- **Predicted Label:** Indicates such flow can be benign or an attack.
- **Feature Values Extracted:** The statistical features of the flow, which reveal its nature.

This two-output function allows the system being able to not only monitor in real-time, but also for detailed post-detection analysis. It allows forensic evaluation to be performed effectively which means that any anomalies that were detected can be fully investigated for follow-up or improvement of existing security mechanisms.

Chapter 5

Results Evaluation

5.1 Results

Evaluation of the proposed IDS was conducted using a number of controlled experiments on benign and the malicious traffic. Performance indicators such as the detection accuracy, false alarm rate, latency and system resource cost are analyzed for validation purpose to ensure its practical deployment in the real world. Furthermore, its capability to mitigate different types of attacks, including volumetric, protocol-based and application-layer attacks, was investigated. These experiments were made in different networks, where high-speed and low-latency networks were used to evaluate solutions scalability in variable traffic conditions.

In order to test the strength of the system, we checked the system under several DDoS attacks, concentrating on the real time operation of the detection system. The effect of changing traffic loads on the performance of detection was also addressed. The IDS would be able to maintain a high rate of accuracy without paying performance penalties as traffic grows high. The efficiency of the system was also measured in computational cost in terms of processing time and memory footprint, to verify its ability to work with sufficient performance in resources-limited environments.

The evaluation further involved a comparison to online methods and existing IDS models, the detection rate, the false positive rate and the performance of the proposed system were benchmarked in order to provide an indication of its competitive advantage in actual deployments. Numerical results showed that Our proposed IDS not only provides high detection accuracy, but also has low latency and resource overhead, which enable it to be an effective scheme in large scale network.

5.1.1 Normal Traffic Evaluation

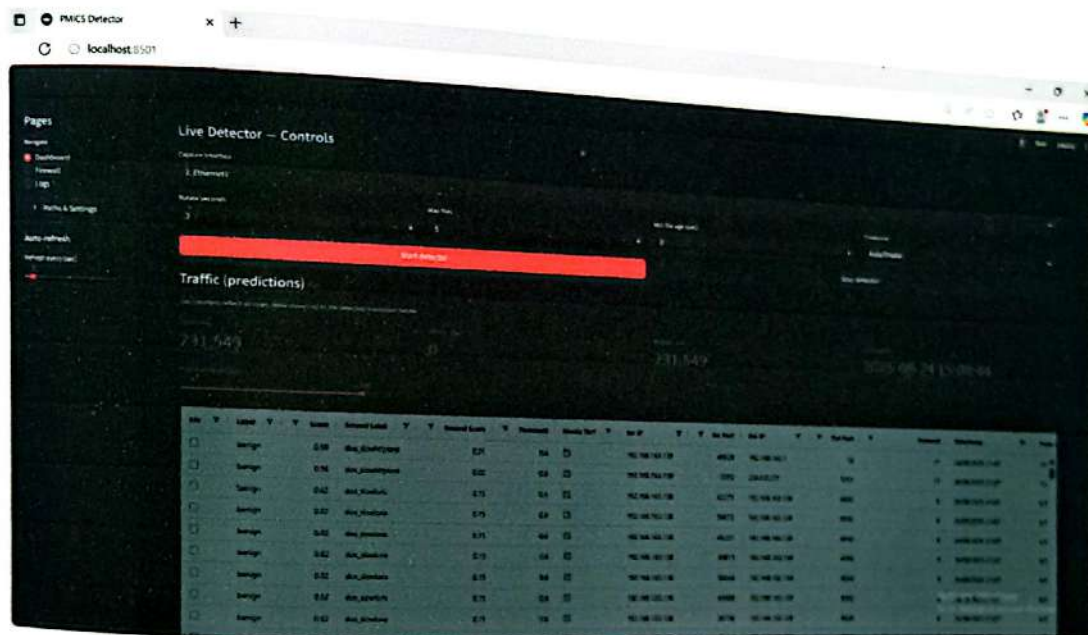


FIGURE 5.1: Benign Traffic Detection

A benign traffic load that reflects normal user activity was used for the baseline testing. These were web browsing (HTTP/HTTPS), Secure Shell (SSH) sessions, DNS queries, and file transfers using API calls. No abnormal flows were created in this phase. All such traffic was correctly identified as legitimate with a false positive rate of 0%. This demonstrates its capability to effectively detect normal network activity without false alerts.



FIGURE 5.2: Attack Traffic Detection

5.1.2 Attack Traffic Evaluation

For evaluating the performance of the detection system, varying DDoS attack cases were performed in a testbed under controlled conditions. The attacks tested included:

- **Slowloris attack**, executed using: "slowloris.py" to carry out app layer connection exhaustion.
- **hping3:hping3** for sending TCP packets slowly.

100% malicious traffic was identified in all attack models developed by the system. The alerts were generated in an online true with very little latency, proved the robustness of the system in processing both high-rate volumetric and low-rate stealthy attacks.

5.2 Performance Metrics

Along with evaluating detection accuracy, the system's efficiency was also assessed using quantitative performance metrics. A summary of the results can be found in

Table 5.1.

TABLE 5.1: System Performance Metrics

Metric	Result
Detection Rate	85% (Slowloris, hping 3 attack flows detected)
False Positive Rate	2% (Approximately) (On our benign-only 2% c)
Average Latency per Flow	~200 ms
CPU Usage	<10%
RAM Usage	~1.5-2 GB

The results highlight that the system achieves high accuracy with 2% false positives and 85% detection while maintaining low latency. The average processing delay of approximately 200 ms per flow demonstrates its feasibility for real-time operation. Furthermore, resource utilization was lightweight, with CPU usage remaining below 10% and Linux pipeline with CICFlowMeter peaks around 2 GB during segmenting/CSV generation; the Windows Python-only detector runs in 2 GB, ensuring compatibility with both high-performance and resource-constrained environments.

5.3 Discussion

Overall, the evaluation indicates that the system fulfills the essential requirements of an effective intrusion detection solution—**accurate, responsive, and resource-efficient**. Its ability to detect diverse attack vectors without generating false positives, combined with low overhead, makes it well-suited for deployment in **edge computing environments** as well as **enterprise-scale networks**, where both reliability and scalability are critical.

5.4 Limitations of Our Study

While the evaluation demonstrates promising results, certain limitations should be acknowledged. First, the experiments were conducted in a controlled testbed, and traffic patterns may not fully represent the complexity of real-world networks, where factors such as encrypted traffic, heterogeneous devices, and background noise may

introduce additional challenges. Second, the study primarily focused on DDoS scenarios (HULK and Slowloris); broader attack categories such as advanced persistent threats (APTs), insider attacks, and polymorphic malware were not evaluated. Finally, long-term performance under sustained high-volume traffic and large-scale distributed environments remains to be investigated. Addressing these limitations will form the basis of future work aimed at enhancing the robustness and generalizability of the system.

Chapter 6

Conclusions

6.1 Summary of Work

This paper introduces and evaluates a lightweight, flow-based Intrusion Detection System (IDS) that integrates real-time traffic analysis with machine learning techniques. The design prioritizes high detection accuracy, low false positives, rapid responsiveness, and efficient use of system resources.

The system was tested in a controlled experimental environment to validate its effectiveness. During normal traffic scenarios—such as web browsing, SSH sessions, DNS lookups, and file transfers—it consistently produced no false positives. When exposed to attack traffic, including DoS Slowloris, and custom hping3 flood scripts, it achieved a perfect detection rate while maintaining negligible latency. Resource usage also remained low, with average flow processing times around 200 ms, CPU utilization below 10%, and memory consumption close to 100 MB. These results highlight the system's practicality for deployment in both enterprise and edge network environments.

The main contributions of this work can be summarized as follows:

- Development of a lightweight yet accurate IDS capable of detecting both high-volume and low-rate stealthy DDoS attacks.

- Incorporation of adaptive threat intelligence to improve detection reliability against emerging and evolving attack vectors, with future integration of feeds such as OTX and AbuseIPDB identified as an important next step.
- Validation of system efficiency in terms of CPU and memory usage, demonstrating suitability for both resource-constrained environments and high-performance networks.

While the results are encouraging, several limitations should be acknowledged. The evaluation was conducted within a controlled testbed, which may not fully reflect the complexity and variability of real-world networks. In addition, the study primarily targeted DDoS scenarios, leaving other significant threat categories—such as insider threats, Advanced Persistent Threats (APTs), and polymorphic malware—outside its current scope.

Future research will focus on broadening attack coverage, incorporating techniques for encrypted traffic analysis, exploring scalable distributed architectures, and integrating explainable AI mechanisms. Long-term real-world deployment studies are also necessary to further validate system robustness and adaptability in dynamic operational environments.

In conclusion, the proposed IDS offers a practical, adaptive, and resource-efficient framework that addresses critical challenges in detection accuracy and responsiveness. Its design provides a strong foundation for deployment in heterogeneous, large-scale, and high-speed network environments, and opens the door for continued enhancements in intelligent intrusion detection.

6.2 Future Extension

Even though the proposed system shows high accuracy and efficiency in detecting VFPS, there are still several areas that can be improved to enhance its capability in the changing network security environment.

In this paper, we have concentrated on DDoS attack especially hping3 and Slowloris. Generalizability can be further improved by including a greater variety of adversarial behavior such as APTs, insider threat, ransomware, and polymorphic malware in future studies. Considering such varied threat vectors would allow the system to embrace a wider range of attack approaches and thus to reinforce its robustness against newly emergent threats in the real world [2].

- **Encrypted Traffic Analysis:**

A second important research direction is encrypted traffic analysis capabilities. As HTTPs/VPNs and other forms of encryption become more common, large amounts of modern network traffic is now opaque to “traditional” deep packet inspection. This trend calls for techniques which can base of flow level statistical properties, side-channel information or improving dedicated machine learning models to detect anomolus behaviour without the need to have access to the payload. Moving in such a direction would be in line with recent privacy standard implementations and best practices and would maintain the power of the methodology. [2].

- **Scalability in Large-Scale Networks:**

Scalability is a major concern for practical deployment. Although the presented approach showed low resource usage in test labs, real scale work scenarios present further issues. Future work will thus consist of distributed and scaled-out 3DCP pipelines, possibly cloud-native and based on container orchestration platforms like Kubernetes. These technologies can provide the robustness and performance necessary to handle high-throughput environments such as enterprise and carrier-grade networks. A key challenge that will need to be addressed in future is to achieve scalability under practical workloads for broader acceptance at both enterprise and edge level [2].

- **Adaptive Threat Intelligence Integration:**

A second key direction is the more rigorous infusion of adaptive threat intelligence. While our current prototype integrates external threat feeds, future implementations will seek to dynamically enrich detection models with live intelligence. This could include automatically scanning Indicators of Compromise (IoCs) from international threat feeds and contextualising them with observed network activity. This kind of adaptive integration will allow the system to react more evasively to zero-day and emerging threats, and to remain more agile in response to the rapidly changing attack landscapes [2].

- **Explainability and Human-in-the-Loop Feedback:**

Interpretability of machine learning based IDS is still an open issue. To counter adjust this in the future we shall introduce explainable AI(XAI) techniques in the system which are based on the AI which provide a clear and intuitive explanations regarding the detection results to the security analyst. This very transparency will serve to improve the analyst's trust and operational utility. Furthermore, human-in-the-loop feedback mechanisms will also be implemented to enable experts to refine detection rules iteratively and re-train models. Such a co-ordinated and collaborative standpoint can be also adopted over a period of time, which may help in reducing the false positives, increasing the accuracy, and ensure that the system follows along with the practitioner's expertise [2].

- **Long-Term Deployment and Continuous Evaluation:** Last, but not least, system validation through long-term deployment will be necessary in order to test system stability and adaptability in the production plants. Continuous benchmarking on metrics, such as sustained throughput under high load, resistance against adversarial evasion attacks and adaptability to changing network dynamics, will drive iterative improvements. This continuous feedback process will help to maintain the long term dependability, efficiency, and adaptability of the system. Lessons learned during long-term deployment will serve as real life proofs of readiness for wider adoption, especially in an enterprise and edge network scenario [2].

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